Piotr Sikora

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Mathematics-focused Data Science student at Lodz University of Technology with strong hands-on experience in Python, R, SQL, and statistical modelling, including time series forecasting and machine learning pipelines. Familiar with the banking environment through direct customer-facing experience at a major commercial bank. Eager to apply quantitative and analytical skills to risk model development, regulatory calculations, and data quality analysis in a professional finance setting.

Education

Lodz University of Technology — Lodz, Poland
Mathematical Methods in Data Analysis, fourth year, 4.71 GPA. — 2023 - present
Rector's Scholarship - awarded to the top 10% of students.
Erasmus program semester in University of Ioannina, Greece.

ZSP 10 im. Jana Szczepanika — Lodz, Poland
Profil Technik Informatyk — 2018 - 2022
Matura: Rozszerzona Matematyka - 80%; Rozszerzona Fizyka - 60%; Rozszerzony Angielski - 96%.

Skills & Interests

Technical: Python, SQL, R, Git, PySpark MLlib, PyTorch, scikit-learn, transformers, ggplot2, R Markdown
Data & Engineering: Apache Spark, Apache Kafka, Docker, machine learning, time series, NLP, knowledge distillation, MS Office
Language: English (B2+), Polish (Native)
Interests: Music, Movies, Team Sports, Martial Arts

Projects

NBA Player Salary Prediction - Real-Time ML Pipeline · Python, Kafka, Spark, Flask, Docker

- Designed an end-to-end ML system ingesting streaming data via Apache Kafka, processing it at scale with Apache Spark, and exposing model comparisons through a Flask API.
- Fully containerised in Docker, demonstrating production-oriented engineering discipline and attention to pipeline reliability.

Financial Market Forecasting with Time Series Models · R, Python

- Conducted full forecasting pipeline on two index fund datasets: raw data ingestion, stationarity testing (ADF and KPSS), decomposition, and comparative model evaluation using ARIMA and ETS.
- Produced forecast visualizations and analytical commentary, closely mirroring the data quality analysis and risk reporting workflow described in the role.

R Data Visualization Reports - Salaries & Consumer Behavior · R, ggplot2, R Markdown

- Produced two fully reproducible EDA reports on real-world datasets, uncovering salary trends across geographies and roles, and seasonal consumer behavior patterns.

SQL Data Exploration · SQL, Python

- Wrote queries across relational datasets using joins, window functions, and subqueries; integrated outputs into Python analysis workflows.

Emotion Analysis via Knowledge Distillation · Python, PyTorch, DistilBERT

- Fine-tuned a DistilBERT student model on an unlabeled, sensitive out-of-domain dataset using soft-label distillation (KL divergence) from a teacher model.
- Benchmarked against a hard-label baseline, showing rigorous experimental methodology and model evaluation skills.
- Directly relevant to LLM-based workflows: transformer fine-tuning via soft-label distillation mirrors the model adaptation techniques used in modern generative AI and agentic systems.

Work Experience

mBank S.A. Lodz, Poland
Customer Service Representative 06/2022 - 09/2022

- Handled high volumes of customer queries relating to banking products, accounts, and financial processes, developing a strong working understanding of retail banking operations.
- Communicated complex regulatory and product information clearly to customers, building skills in accuracy, compliance awareness, and professional communication.

Achievements & Activities

Conference Speaker Lodz, Poland
MathUp 2025, Lodz University of Technology 04/2025

- Delivered a lecture in English titled *"Making Sense of Higher Dimensions"* on dimensionality reduction techniques.
- Awarded Best Lecture in English by the Director of the International Faculty of Engineering.

National-Level Rugby Athlete Lodz, Poland
KS Budowlani Lodz 2012 - 2023

- Selected for the Polish national team; won 2 youth championships and secured 2 national league promotions, including to the highest division of Polish rugby.
- Developed the discipline, composure under pressure, and team-oriented mindset directly transferable to fast-paced, collaborative work environments.